

POLI 110 Tutorial - Week 1

Introduction to Scientific Methods for Political Analysis

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Course Requirements

● Course Evaluation

- Midterm Exam (25%) - February 26th
- Final Exam (35%) - TBD by registrar
- 5 Homework Assignments (25% total, 5% each)
- Tutorial Participation (15%)

● Homework Assignments

- Based on Vox.com "explainer" articles
- Choose from provided list of topics
- Apply course concepts to real-world cases
- Due dates posted on Canvas
- Late submissions: 50% penalty within 24h, no credit after

● Tutorial Participation

- Weekly attendance required
- Active participation in discussions
- Quality of contributions matters

Part 1: What is Social Science?

• Identifying Meaningful Patterns

- Social science seeks to understand patterns in society
- Focus on systematic and recurring phenomena
- Not just individual cases, but broader trends

• Political Science Context

- Patterns in distribution of political power
- How these patterns affect individuals and groups
- Impact on political behavior and outcomes

Examples of Political Patterns

• **Voting Behavior**

- Income levels and voting preferences
- Education and political participation
- Geographic patterns in electoral outcomes

• **Power Distribution**

- Institutional structures and decision-making
- Resource allocation across regions
- Representation in political bodies

Part 2: Challenges in Research Methods

- **Measurement Challenges**

- How do we measure abstract concepts?
- Example: How do we measure "democracy"?
- Potential for measurement error

- **Sampling Issues**

- Who do we study?
- How many cases do we need?
- Selection bias concerns

- **Measurement Errors**

- Imprecise definitions
- Poor operationalization
- Unreliable data collection

- **Sampling Problems**

- Non-representative samples
- Too small sample size
- Missing important cases

Part 3: Causation and Counterfactuals

• **The Fundamental Problem of Causal Inference**

- We need to compare what happened with what would have happened
- But we can never observe both outcomes for the same unit
- Example: Can't observe same person voting both with and without college education

• **Counterfactual Thinking**

- What would have happened in the absence of treatment?
- The unobservable alternative scenario
- Key to understanding true causal effects

Dealing with Selection Bias

- **The Problem**

- Units self-select into treatment
- Systematic differences between groups
- Biased estimates of causal effects

- **Solutions**

- Random assignment when possible
- Control for confounding variables
- Natural experiments
- Matching methods

- **Sources of Endogeneity**

- Reverse causation
- Omitted variables
- Simultaneous causation

- **Examples in Political Science**

- Democracy and economic development
- Political participation and civic education
- Institution design and political outcomes

- **Addressing Endogeneity**

- Instrumental variables
- Fixed effects
- Research design strategies

- **Requirements for Causation**

- Temporal sequence (cause before effect)
- Theoretical mechanism
- Ruling out alternative explanations

- **Example in Political Science**

- Correlation: Higher education and voter turnout
- But is education causing participation?
- Or are there other factors involved?

Discussion Questions

Consider these research methodology questions:

- 1 How can we accurately define and measure our key concepts and variables when studying the relationship between negative emotions and radical party voting?
- 2 What potentially confounding or mediating factors should we control for to establish a valid statistical relationship?
- 3 What research methods and data collection approaches would be most appropriate and why?